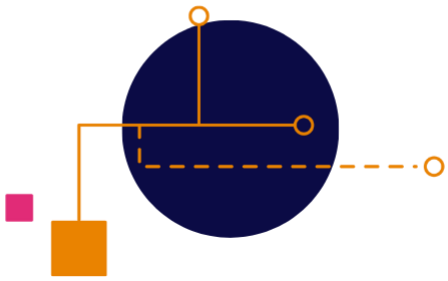
Rockborne
A HARNHAM GROUP COMPANY

Python Project - Crime data Pipeline & Reporting Dataset



Context

You are part of a **Police Force Analytics Unit**, supporting leadership teams who need reliable data to monitor crime patterns across regions and over time.

The data volume is large and operational in nature, requiring scalable, robust data handling practices. In this phase, you are working as a data engineer within the analytics team.

Your responsibility is to prepare high quality, well documented data assets that analysts will ingest into a Business Intelligence tool to support monitoring, comparison, and prioritisation across police forces.

Data

- Primary dataset (required): You must choose 4 police forces to work with
    - UK Police Crime Data: https://data.police.uk/data/
- Enrichment datasets (required): You must integrate **between 1 and 3 additional datasets** to enrich the crime data. These datasets should provide contextual or normalisation information that supports operational comparisons, such as (but not limited to):
    - Population data (e.g. population by police force or region)
    - Deprivation or socioeconomic indicators
    - Employment, housing, or demographic metrics

Constraints:

- All enrichment datasets must be:
    - Publicly available
    - Processed **in Python**
- Each dataset must add **clear analytical value** and be justifiable for operational reporting
- You must document:
    - Why each dataset was chosen
    - How it was processed

- Any assumptions or limitations introduced

To make the dataset suitable for operational reporting, you must perform data aggregation as part of your pipeline. The final output should not remain at individual crime-record level; instead, it should be aggregated to a single, clearly defined reporting grain (e.g. police force × month × crime category).

Your aggregation should produce BI-ready measures such as crime counts and, where relevant, normalised rates (e.g. crimes per 1,000 residents). Ensure the chosen grain aligns with your enrichment data and document any assumptions used to align time periods or geographies.

Intended Use of the Dataset

The analytics team has asked you to prepare a reporting ready dataset that enables them to answer the following questions in a BI tool:

1. How is overall crime changing over time for each police force?
2. What crime types drive crime volume, and how does the mix change over time?
3. Are there seasonal patterns or anomalous spikes at force or category level?
4. How do selected police forces compare head-to-head on key metrics?
5. How do forces compare on normalised crime rates (e.g. per 1,000 residents)?
6. How does crime data relate to population and socioeconomic context?
7. Do any areas show repeat seasonal patterns?
8. Are there geographic clusters (hotspots) of specific crime categories, and how do these clusters differ by police force? (optional)

Important:
Your responsibility is to ensure the **data structure, grain, and quality** allow these questions to be answered downstream. You are not expected to analyse or explain trends in this phase.

Required Components (Deliverables)

1. Statement of Work

2. Data Engineering Pipeline (Jupyter Notebook- this should be one notebook)

Your notebook(s) must be structured as a clear, repeatable pipeline with the following stages:

Ingestion layer

- Read data in batches or chunks (e.g. by file, time period, or police force)
- Avoid loading the full dataset into memory in a single step without justification

Cleaning & validation layer

- Data quality assessment
- Handle missing values, duplicates, and inconsistent category values
- Implement explicit validation checks, including:
    - Row counts before and after transformations
    - Duplicate checks at the intended reporting grain

Feature engineering & transformation

- Derive time attributes (e.g. year, month)
- Standardise crime category fields
- Join 1–3 enrichment datasets, ensuring:
    - Compatible grain
    - No unintentional “many to many” joins
    - Clear handling of temporal mismatches (e.g. annual population vs monthly crime)

Aggregation for reporting

- Aggregate the data to a single, consistent reporting grain
- Prepare metrics required for BI consumption (e.g. crime counts, rates)

Export layer

- Export one final reporting ready dataset
- Ensure the pipeline runs end to end from raw data to final output
- Validate the processes by making a generic analysis on high level statistics (e.g. total rows, number of regions, etc.)

3. Reporting Dataset Requirements (for Power BI). You must produce one reporting ready CSV file that:

- Has a clearly documented grain (e.g. police force × time period × crime category)
- Includes aggregated measures required for BI reporting
- Contains clean, consistent categorical values
- Includes normalised metrics where applicable (e.g. per capita rates)
- Has validation checks confirming:
    - No unintended duplication at the reporting grain
    - No missing values in required reporting fields

4. Workflow diagram showing ingestion → transformation → output

5. Data Dictionary for the reporting dataset

    - Column definitions
    - Data types
    - Grain
    - Sources
    - Calculations and assumptions

Phase 1 – Data Engineering Marking Scheme

Topic	Description	Marks
Data Pipeline Design, Ingestion and Enrichment	Assesses whether the solution is a clear, reusable data engineering pipeline with distinct stages from ingestion to export. It covers responsible batch, chunked, or iterative handling of UK Police crime data, plus relevant enrichment datasets processed in Python and joined at compatible grains with appropriate time alignment, validated joins, and no unintended duplication or misleading outputs.	**1–2: Pipeline is ad hoc, fragile, or poorly structured. Ingestion is naïve, enrichment is weak or poorly joined, and there is limited evidence of scalability, grain control, or duplication checks.** **3–4: Pipeline is mostly structured, with some staged processing and batch or iterative ingestion. Enrichment is relevant, but documentation, time alignment, join validation, or assumptions may be incomplete.** **5: Clear, reusable end-to-end pipeline with strong staging and scalable ingestion. Enrichment datasets are well chosen, processed in Python, correctly aligned by grain and time, and validated to prevent unintended duplication.**

Data Quality, Validation & Correctness	Assesses how systematically data quality is checked, validated, and preserved throughout the pipeline.	**1–2:** Minimal or no data validation. Duplicate records, missing values, or integrity issues are likely undiscovered or undocumented. **3–4:** Validation checks exist but are incomplete or inconsistently applied. Some issues are identified but not fully addressed. **5:** Robust, explicit validation checks (e.g. duplicates at reporting grain, null checks, row count reconciliation, category validation). Data integrity is clearly enforced and documented.
Reporting Dataset, Documentation & Handoff Readiness	Assesses the quality of the final reporting dataset and how well it is documented for downstream BI and analytics use.	**1–2:** Reporting dataset is incomplete or unclear. Data dictionary and documentation are missing, inaccurate, or insufficient for downstream handoff. **3–4:** Reporting dataset is mostly BI-ready, with minor issues. Documentation and data dictionary exist but lack detail or clarity in places. **5:** Final dataset is clean, BI-ready, and aggregated to a clearly documented grain. Data dictionary is complete and professional (definitions, types, sources, derivations, assumptions). Workflow diagram clearly explains the pipeline.